SYSTEM AND METHOD FOR COVERT COMMUNICATION USING ARRANGEMENTS OF SATELLITE-OBSERVABLE, DISTINCTIVELY COLORED OBJECTS

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Title: SYSTEM AND METHOD FOR COVERT COMMUNICATION USING ARRANGEMENTS OF SATELLITE-OBSERVABLE, DISTINCTIVELY COLORED OBJECTS

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Abstract

A method and system for secure, one-way communication that uses no radio frequency (RF) transmissions. The system encodes messages by physically arranging a plurality of large, movable objects on the ground. These objects are characterized by having specific, statistically rare colors, often protected by trademark, which allows them to be uniquely and accurately identified in satellite imagery against environmental clutter. A message is composed by arranging a set of these objects in a specific spatial configuration within a designated area, such as an industrial equipment rental yard. An orbiting satellite captures an image of the area, and a remote computer vision system decodes the message by analyzing the spatial arrangement, orientation, and color of the objects. The system allows for covert

communication of status updates or simple messages from an operative on the ground to a remote command center without creating any interceptible electronic signals.

Background of the Invention

Covert communication is a cornerstone of intelligence and military operations. Traditional methods rely on radio frequency (RF) transmissions, which, even when encrypted, are subject to detection and interception. An adversary who can detect the mere presence of a transmission can locate the sender, even if the message cannot be decoded. This creates a significant operational security risk.

There is a long-standing need for communication methods that produce no electronic emissions and are therefore "invisible" to signals intelligence (SIGINT). While visual signaling methods like semaphore or signal lamps exist, they are limited by line-of-sight and are not suitable for long-distance communication to a remote, non-visible party. The present invention addresses this need by leveraging modern satellite imaging and computer vision technology to create a high-latency but extremely secure one-way communication channel.

Summary of the Invention

The present invention is a system for covert communication that uses arrangements of physical objects on the ground as a medium for encoding information. A sending operative, located at a site such as an equipment rental yard, composes a message by arranging several large, movable objects, such as construction vehicles (e.g., excavators, tractors) or shipping containers.

A key feature of the invention is that the objects used are painted in specific, highly distinctive, and often trademark-protected colors (e.g., "Caterpillar Yellow," "John Deere Green"). Because these colors are legally protected and thus statistically rare, they provide a high signal-to-noise ratio when viewed in satellite imagery. This allows a remote computer vision system to easily identify the objects of interest and ignore background clutter.

Information is encoded in multiple dimensions: the specific combination of colored objects used, their spatial arrangement relative to each other, and, for objects with an asymmetrical shape, their orientation. For example, a simple binary status ("target present") could be encoded by the presence of a single, symmetrically-shaped object (e.g., a round logo on a truck roof). A more complex message could be encoded by the orientation of an asymmetrical object (e.g., an excavator pointing north to signify "proceed," or south to signify "abort").

An imaging satellite periodically captures images of the designated signaling area. The images are downlinked to a secure processing center, where a computer vision algorithm analyzes the images, identifies the colored objects, and decodes the message based on their arrangement and orientation. This creates a broadcast-only, non-RF communication link from the ground to the remote center.

Brief Description of the Drawings

FIG. 1 is a diagram illustrating the overall system, showing movable, colored objects on the ground being imaged by a satellite.

FIG. 2 is a conceptual diagram illustrating how a symmetrical object can be used to encode a binary status.

FIG. 3 is a conceptual diagram illustrating how an asymmetrical object can be used to encode a multi-state message via its orientation.

Detailed Description of the Invention

Referring to **FIG. 1**, the system comprises a ground segment and a space segment. The ground segment consists of a set of movable, satellite-observable objects (101), such as construction vehicles, located in a designated signaling area (102), for example, an equipment yard. These objects are notable for their distinctive, often trademark-protected, colors.

A ground operative (not shown) arranges the objects (101) into a specific spatial configuration to encode a message. An orbiting satellite (103) with an imaging payload captures an image of the signaling area (102). The image data is downlinked to a remote Processing Center (104). At the center, a Computer Vision System (105) analyzes the imagery, identifies the objects (101) based on their unique colors and shapes, and decodes the message based on their pre-defined arrangement. The decoded message (106) is then relayed to the intended recipient.

The method of encoding can vary in complexity. **FIG. 2** illustrates a simple binary encoding scheme. A large, movable object, such as a truck (201), has a symmetrical, distinctively colored logo (202) on its roof. By parking this truck near a target location (203), an operative can signal the binary status "target is present" to the satellite imaging system. The presence or absence of the truck at the location is the entirety of the message.

FIG. 3 illustrates a more complex encoding scheme using an asymmetrical object, such

as an excavator (301). Because the object is asymmetrical, its orientation is observable from satellite imagery. An operative can park the excavator with its arm pointing in one of several pre-defined directions (e.g., North (302), East (303), South (304), West (305)). Each direction corresponds to a different state in a pre-arranged codebook (e.g., North = "Proceed," South = "Abort," East = "Standby"). This allows for the transmission of a multi-state message using a single object, purely through the manipulation of its physical orientation. By combining multiple objects of different colors and orientations, a richer and more complex message can be composed and transmitted without any electronic emissions.

Claims

What is claimed is:

1. A system for covert communication, comprising:
    - a plurality of movable, physical ground objects, wherein each object is colored with a specific, statistically rare color;
    - an imaging satellite configured to periodically capture images of a designated geographical area containing said ground objects; and
    - a remote data processing system configured to:
        - receive said images from the satellite;
        - execute a computer vision algorithm to identify the ground objects within the images based on their statistically rare color; and
        - decode a message based on a pre-defined meaning associated with the spatial arrangement of the identified ground objects.
2. The system of claim 1, wherein the statistically rare color is a trademark-protected color.
3. The system of claim 1, wherein the ground objects are construction vehicles or shipping containers.
4. The system of claim 1, wherein at least one of the ground objects has an asymmetrical shape, and wherein the message is further decoded based on the rotational orientation of said asymmetrical object.
5. A method for covert, non-RF communication, comprising the steps of:

- Physically arranging one or more movable ground objects of a distinctive color into a specific spatial configuration within a designated area;
- Capturing, via an orbiting satellite, a digital image of said designated area;
- Transmitting the image to a remote processing center; and
- At the processing center, automatically analyzing the image to identify said objects and decode a message based on their spatial configuration.

6. The method of claim 5, wherein the distinctive color is a trademark-protected color, and wherein the step of analyzing the image uses said color to distinguish the objects from environmental clutter.

7. The method of claim 5, wherein at least one of the objects is asymmetrical, and further comprising the step of encoding information in the rotational orientation of the asymmetrical object.

8. The method of claim 7, wherein the decoded message is a status update corresponding to one of a plurality of pre-defined states, each state being associated with a specific rotational orientation.

Drawings

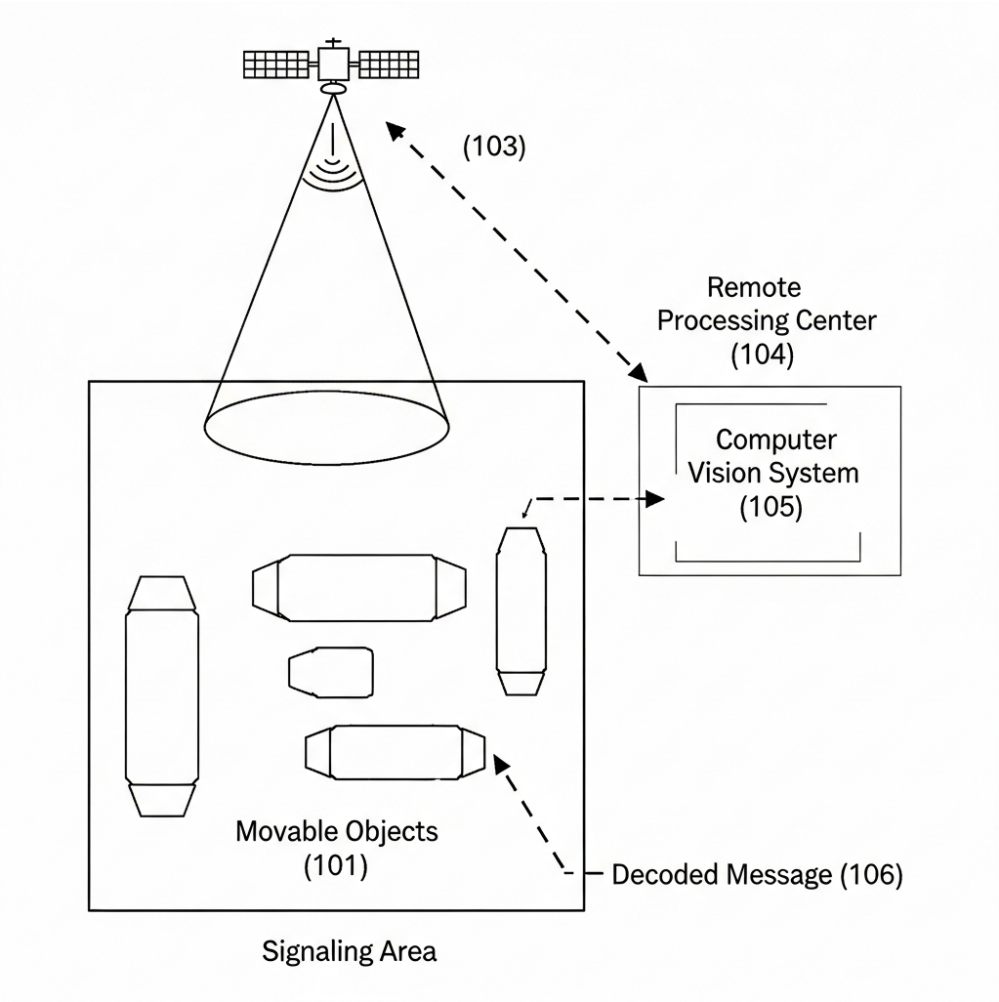


Figure 1: Overall system diagram.

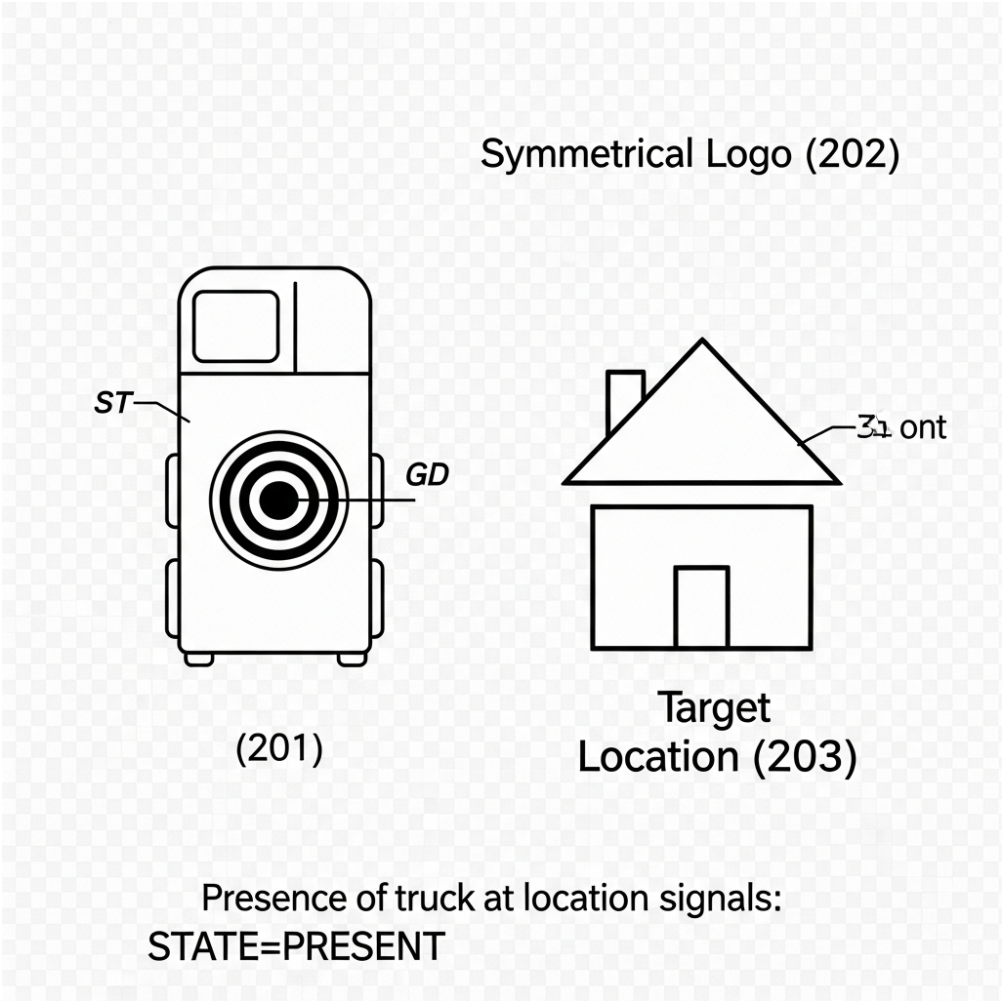


Figure 2: Binary encoding with a symmetrical object.

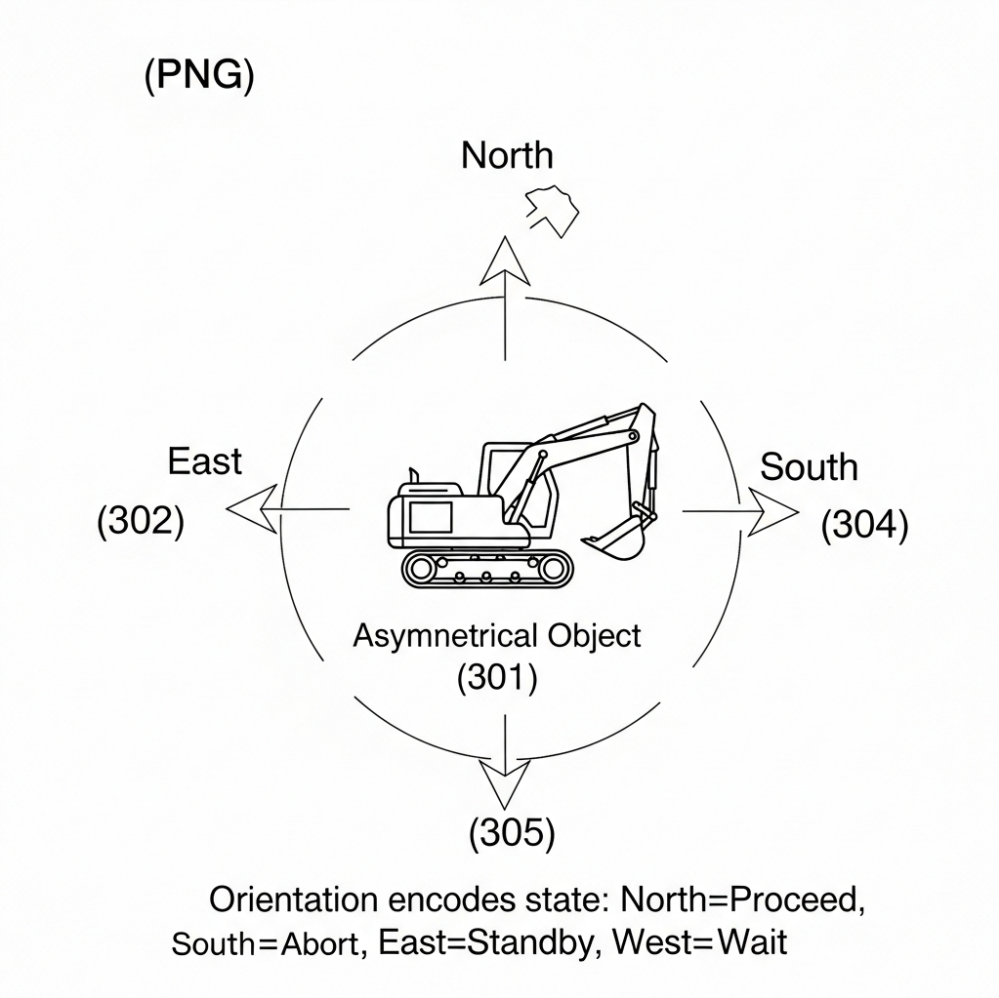


Figure 3: Multi-state encoding with an asymmetrical object.